ENERGY POLICY

Why Nobody Blinks: The Nash Equilibrium Inside India's E20 Standoff

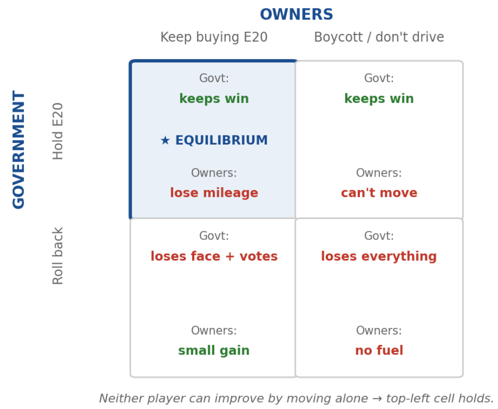
The government won't roll back ethanol-blended petrol. Owners won't stop buying it. That's not paralysis - it's game theory.

In 2014, ethanol made up just 1.5% of India's petrol. By 2025 it was 20% - the fastest ramp-up of any ethanol-blending nation, saving a claimed Rs 1.06 trillion in crude imports and 54.4 million tonnes of emissions. Car owners see it differently: a June 2026 survey of 44,000-plus owners found 66% of pre-2023 cars losing over 10% of their mileage. The anger peaked when the government's own attorney general called E20 an “experiment” in the Supreme Court. Officials called it a misunderstanding. The backlash stuck.

And yet nothing has moved. E20 still owns the pumps. This is where game theory beats outrage: a **Nash equilibrium** is a standoff where nobody can improve their position by moving alone - so nobody moves. It doesn't mean everyone is unhappy. It means the winners have no reason to move, and the losers have no way to.

The government is a winner and can't reverse E20 without admitting a flagship policy failed. Farmers are the bigger winner - the roughly Rs 40,000 crore a year flowing to them for ethanol is a vote bank no government unwinds lightly, and they are the real reason the top row stays locked. Owners are the losers, but they can't exit either: there's no E10 pump left to drive to, so complaining costs less than a car that doesn't move.

> ***“Nobody can improve their outcome by switching strategy alone - so the stalemate holds.”***



The payoff grid: every alternative to the top-left cell leaves at least one player worse off, so neither side defects.

There's a third cost nobody at the table is paying - and it may be what keeps the equilibrium stable. Sustaining 20% blending has diverted sugarcane, maize and rice into fuel, and researchers note ethanol from food crops can lose much of its climate edge once farming and processing emissions are counted. That bill falls on food supply and land use, borne by people who aren't players in the game - which is precisely why it exerts no pressure on it.

And the government isn't just holding this line - it's building past it. The roadmap runs only to October 2026, an Inter-Ministerial Committee review is still pending, yet regulators have already cleared an E30 fuel standard and launched E85 pilots at 48 stations. The reply to a backlash over 20% ethanol is quiet preparation for 30% and 85%.

So E20 sits exactly where a Nash equilibrium predicts: not scrapped, not vindicated, just persisting. What breaks it won't be public anger - it's a new player entering the game: a hard finding of engine damage, the pending committee report, or an election that shifts the farmer math. Until then, blinking first costs more than staying put.

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